

ENGR. MUSTAKIM BILLAH RAFI

System Design and Backend Engineer

rafisharkar144@gmail.com | +880 1905493909 | linkedin.com/in/rafi-sharkar | github.com/Rafi-Sharkar | rafisharkar.com

SUMMARY

System Design and Backend Engineer with 4 years of progressive experience spanning frontend, fullstack, and backend engineering, including 2 years at Softvence IT Ltd where I advanced from Backend Developer to System Design and Backend Engineer. Currently maintain two live client-facing platforms with **99.9% uptime**, and specialize in designing scalable systems using **Node.js, NestJS, TypeScript, and PostgreSQL**. Broad background covers React/Next.js frontend engineering, fullstack delivery, RESTful and event-driven API design, WebSocket real-time systems, Stripe payment integration, Redis caching, and cloud deployment via **AWS, Docker, and CI/CD pipelines**. Published researcher (Springer, 2024) with a BSc in Computer Science and Engineering from Independent University Bangladesh.

SKILLS

Languages: TypeScript, JavaScript, C#, Python, Java, SQL

Frameworks and Databases: Node.js, NestJS, Next.js, React, Express.js, PostgreSQL, MongoDB, Redis, Prisma ORM

APIs and Authentication: REST API, GraphQL, gRPC, WebSocket, WebRTC, JWT, OAuth2, RBAC, Swagger/OpenAPI

DevOps and Tools: Git, GitHub Actions, AWS, Docker, Kubernetes, CI/CD Pipelines, Linux, System Design, Data Structures and Algorithms

Practices: Microservices Architecture, System Design, Agile/Scrum, Test-Driven Development (TDD), Unit Testing, Integration Testing, End-to-End Testing

Architecture and Reliability: Cloud Architecture (AWS), API Gateway, Load Balancing, Application Monitoring, Logging, Security Best Practices

Frontend (early career): React.js, Next.js, HTML5, CSS3/SASS, Redux, Responsive UI Design

WORK EXPERIENCE

System Design and Backend Engineer | Sep 2025 – Present

Softvence IT Ltd | Dhaka, Bangladesh

Key Skills: System Design, Microservices Architecture, Node.js, NestJS, PostgreSQL, Redis, AWS, Docker, API Gateway, Load Balancing, JWT/OAuth2, RBAC

- Promoted from Backend Developer to lead system design for two live production platforms — **childcareregister.com** and **readytoworkapp.com** — serving **500+ active users** with **99.9% uptime** on AWS.
- Own end-to-end architecture decisions: service boundaries, database schema design, caching strategy (Redis), and API gateway/load-balancing configuration for scalable multi-tenant delivery.
- Lead design and code review for **30+ REST API endpoints** secured with JWT, OAuth2, and multi-tier RBAC, reducing unauthorized access incidents to zero.
- Drive CI/CD and containerization strategy (Docker, GitHub Actions), reducing deployment time by **40%** and enabling zero-downtime releases.
- Mentor junior engineers and coordinate with frontend, UI/UX, and AI teams in an Agile/Scrum workflow, delivering **3+ sprint releases** on schedule.

Backend Developer | Sep 2024 – Sep 2025

Softvence IT Ltd | Dhaka, Bangladesh

Key Skills: Node.js, NestJS, TypeScript, PostgreSQL, WebSocket (Socket.io), BullMQ, Stripe API, Jest, Docker, GitHub Actions

- Built and maintained core backend services for **childcareregister.com** and **readytoworkapp.com** using Node.js, NestJS, and PostgreSQL.
- Built real-time **chat and notification system** using Node.js, WebSocket (Socket.io), and BullMQ, cutting notification delivery latency by **approximately 60%** versus polling.
- Integrated **Stripe payment processing** handling end-to-end transaction flows; wrote unit and integration tests using **Jest** achieving **80%+ code coverage**.
- Containerized services with **Docker** and set up initial **CI/CD pipelines** with GitHub Actions.

Fullstack Developer | Sep 2023 – Sep 2024

Metamorphosis | Remote

Key Skills: React.js, Next.js, Node.js, Express.js, PostgreSQL, MongoDB, JWT, RBAC, REST API Design

- Developed and maintained fullstack features across React/Next.js frontends and Node.js/Express backends for client web applications.
- Designed REST APIs and integrated them with PostgreSQL/MongoDB databases, handling authentication (JWT) and role-based access control.
- Collaborated with designers and product owners in an Agile workflow to ship UI components and backend endpoints in the same sprint cycle.

Remote Frontend Developer | Sep 2022 – Sep 2023

Unitechnology | Remote

Key Skills: React.js, Next.js, JavaScript, HTML5, CSS3/SASS, Redux, REST API Integration, Responsive UI Design

- Built responsive, cross-browser user interfaces using React.js and Next.js, translating Figma designs into reusable components.
- Integrated frontend applications with REST APIs and managed application state using Redux/Context API.
- Optimized page load performance and collaborated with backend engineers to refine API contracts.

SELECTED PROJECTS

Poultry Supply Chain Management System | Aug – Oct 2025

Key tech: Node.js, NestJS, Next.js, TypeScript, PostgreSQL, Prisma, JWT Source Code: github.com/Rafi-Sharkar

- Built full-stack platform digitizing end-to-end poultry operations for **3 user roles** (farmer, distributor, admin), reducing manual order processing by **approximately 70%**.
- Designed normalized PostgreSQL schema with **12+ relational tables**, supporting stock tracking, order management, and delivery modules.

Real-Time Chat and Notification Service | 2025

Key tech: Node.js, NestJS, TypeScript, WebSocket (Socket.io), PostgreSQL, JWT Source Code: github.com/Rafi-Sharkar

- Architected WebSocket-powered messaging service with **multi-room chat, read receipts, and push notifications**, tested with Jest to **85% coverage**.
- Designed event-driven architecture with JWT-authenticated Socket.io connections, supporting **concurrent sessions** at scale.

ERP System for Manufacturing Operations | 2024

Key tech: NestJS, Next.js, TypeScript, PostgreSQL, Prisma, Redis, RBAC Source Code: github.com/Rafi-Sharkar

- Designed modular ERP covering **inventory, HR, and finance** modules with role-based dashboards for admin, manager, and staff users.
- Implemented audit-logged transaction workflows and automated low-stock/reorder alerts, reducing manual inventory reconciliation effort.
- Built reporting layer with exportable PDF/Excel summaries and Redis-cached queries for large dataset performance.

Multi-Tenant SaaS Subscription Platform | 2024

Key tech: Node.js, NestJS, Next.js, TypeScript, PostgreSQL, Stripe, Docker Source Code: github.com/Rafi-Sharkar

- Built multi-tenant SaaS boilerplate with organization-level data isolation, tiered subscription plans, and Stripe-based billing/invoicing.
- Implemented usage-based feature gating and admin analytics dashboard for tenant activity and revenue tracking.
- Containerized the platform with Docker for consistent staging/production environments and one-command deployment.

Group Chat and Video Calling App | 2024

Key tech: Node.js, WebSocket (Socket.io), WebRTC, TypeScript, MongoDB, JWT Source Code: github.com/Rafi-Sharkar

- Developed real-time messaging app with **peer-to-peer video calling** via WebRTC, group chat rooms, and typing/read-receipt indicators.
- Optimized signaling server for low-latency connection setup and implemented reconnection handling for unstable networks.

Product Landing Page and Marketing Site | 2023

Key tech: Next.js, React, TypeScript, Tailwind CSS, SEO Optimization Source Code: github.com/Rafi-Sharkar

- Built a fully responsive, SEO-optimized marketing landing page with animated sections, contact form, and analytics integration.
- Achieved fast load times through image optimization, code-splitting, and static site generation (SSG) with Next.js.

Social Networking Platform | 2023

Key tech: Next.js, Node.js, Express.js, MongoDB, JWT, REST API Source Code: github.com/Rafi-Sharkar

- Built a social platform with **user profiles, posts, comments, likes, and a follow system**, with a personalized activity feed.
- Designed a notification service for likes/comments/follows and implemented pagination and infinite scroll for feed performance.

KEY ACHIEVEMENTS

- **Promotion:** Promoted from Backend Developer to System Design and Backend Engineer within 1 year at Softvence IT Ltd, based on architecture and reliability contributions.
- **Production Uptime:** Maintained **99.9% uptime** across two live client-facing systems using AWS, Docker, and automated CI/CD pipelines.
- **Research and Leadership:** Led a team of 4; published Springer paper (2024) on a Supply Chain Management System for Agribusinesses — ERD and relational schema design.
- **Hackathon:** Designed a supply chain prototype at **HackNSU Season 5** for RMG sector optimization.

EDUCATION

BSc in Computer Science and Engineering — Independent University Bangladesh, Dhaka | Sep 2021 – Dec 2025

CERTIFICATIONS

Introduction to Networks — CISCO (L2/L3 configuration, IPv4/IPv6, network security) | 2024